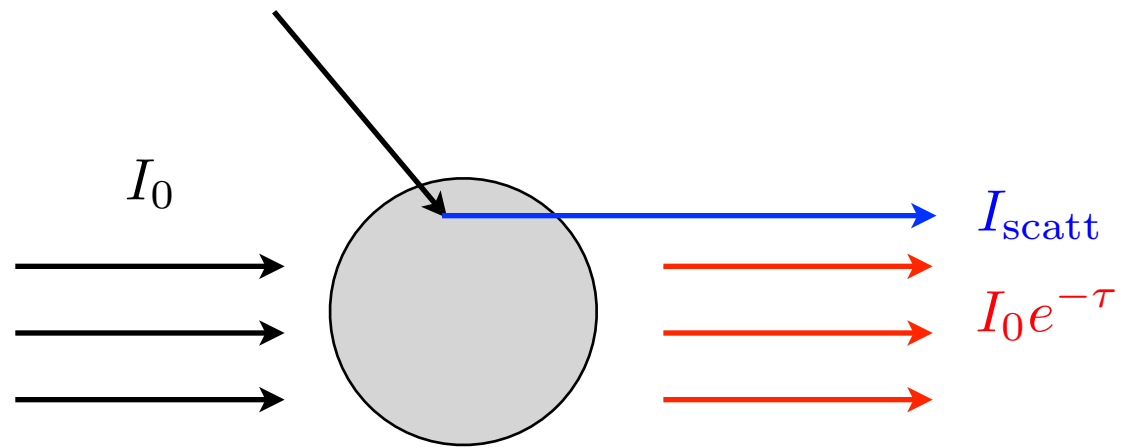
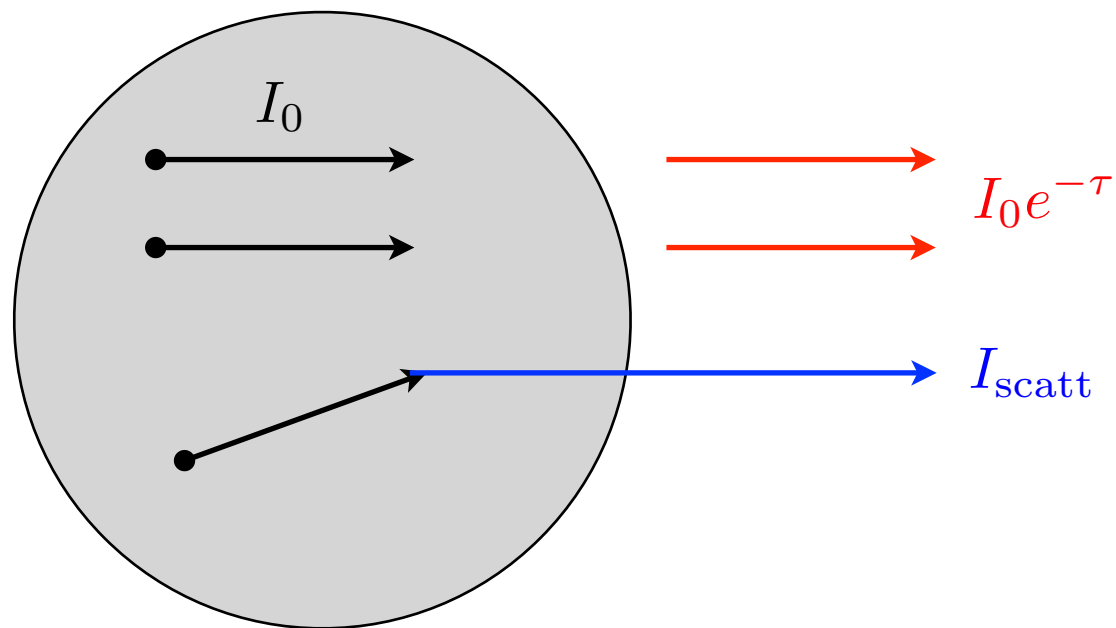


[External Source]



[Internal Source]



Definitions

τ = **optical depth** from the emission site toward the observer

$I_0 e^{-\tau}$ = **direct light** or transmitted light

I_0 = **initial light**, the light which would be observed in the absence of dust along the line of sight

I_{scatt} = **scattered light**

Relation with the Simulation Data

```
a = fits.open('sample_obs.fits.gz')
b = fits.open('sample_obs_tau.fits.gz')
```

```
b[0].data  $\Rightarrow \tau$ 
```

```
a[0].data  $\Rightarrow I_{\text{scatt}}$ 
```

```
a[1].data  $\Rightarrow I_0 e^{-\tau}$ 
```

```
a[2].data  $\Rightarrow I_0$ 
```

The data are all 2D images on a detector plane.

a[2] is saved only if the following condition is specified in the input file:

```
par%save_direct0 = .true.
```